

Blood MERIDIAN: a blockcipher that is not a blockcipher

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AI disclosure. GPT-5.6 Sol (Daybreak Blue) found the results reported here and produced the initial PoC code and manuscript draft. I (JP) manually reviewed the results and code for correctness, and substantially revised the writeup. Another LLM (Gemini Flash) checked the results. This note arose from an afternoon experiment within a broader study of LLM-assisted cryptanalysis, prompted in part by my claim that LLMs wouldn’t break established symmetric designs.¹

MERIDIAN [1] is a 128-bit blockcipher proposed as a lightweight AES alternative. We show that its “Directional Substitution (DS)” layer is not injective by giving an explicit collision. Because encryption begins with key whitening, this yields a full 12-round collision for every key. Consequently, no keyed instance of MERIDIAN is a permutation, so no decryption function can invert encryption on all plaintexts, and its block-cipher and PRP security claims fail. We additionally identify an exact one-round differential that exceeds the claimed bound by a factor 13.37, and explain why the published MILP model does not encode the specified DS layer.

1. The construction

MERIDIAN represents its state as a 4×4 array of bytes. For each position (i, j) , DS computes

$$c_{i,j} = S[i, j-1] \oplus S[i-1, j], \quad r_{i,j} = (c_{i,j} \bmod 7) + 1,$$

$$\text{DS}(S)[i, j] = \text{ROL}_{r_i \cdot i}(\pi(S[i, j])),$$

where π is the AES S-box. A round then applies a linear MD layer, keyed AM, and a round constant. Encryption starts from $S = P \oplus K$ and applies 12 rounds.

2. An explicit DS collision

Let every row of A be 02 29 02 29, and every row of B be 02 46 02 46. Thus

$$A = 0229022902290229022902290229,$$

$$B = 0246024602460246024602460246.$$

Every $c_{i,j}$ in A is $02 \oplus 29 = 2b$, giving rotation 2. Every $c_{i,j}$ in B is $02 \oplus 46 = 44$, giving rotation 6. The relevant AES S-box values are

$$\pi(02) = 77, \quad \pi(29) = a5, \quad \pi(46) = 5a.$$

Direct calculation gives

$$\text{ROL}_2(77) = \text{dd}, \text{ROL}_6(77) = \text{dd},$$

$$\text{ROL}_2(\text{a5}) = 96, \text{ROL}_6(5\text{a}) = 96.$$

Therefore

$$\text{DS}(A) = \text{DS}(B) = \text{dd96dd96dd96dd96dd96dd96dd96}.$$

3. Full-round consequence

For an arbitrary key K , choose the distinct plaintexts

$$P_0 = A \oplus K, \quad P_1 = B \oplus K.$$

Whitening maps P_0 and P_1 back to A and B . Their states become equal after the first DS layer and remain equal under every subsequent deterministic operation. Hence

$$E_K(A \oplus K) = E_K(B \oplus K) \quad \text{for every } K.$$

For $K = 0$, a direct implementation of the formal specification gives the common twelve-round ciphertext

cb2529d5384891c5214b2190846a02f3.

4. Invalid probability bound

Proposition 2 claims a one-round upper bound $(2^{-6})^w 2^{-3w'}$. Put input difference 12 only at byte (0,0) and request DS difference 55 there and zero elsewhere. Then $w = 1$ and $w' = 2$, so the claimed bound is 2^{-12} . Exact enumeration gives

$$p = \frac{804}{65536} \left(\frac{33808}{65536} \right)^2 \approx 2^{-8.258794},$$

which is ≈ 13.37 times larger. The proof incorrectly assigns a 2^{-3} penalty to every changed context. For $\Delta c = 12$, 128 of 256 contexts leave $(c \bmod 7)$ unchanged, so the two rotations coincide with probability at least one half.

For $K = 0$, MD maps this difference to 5500000000000000aa00000055000000, AM is the identity, and the round constant cancels. Thus this is a complete one-round differential with the same probability.

5. Validation note

The paper does not provide 128-bit test vectors. Its worked example uses a 32-byte prototype although Section 3 specifies 16 bytes and the prototype is encrypt-only. Our reproducer follows the formal 128-bit equations. The collision is independently checkable from the AES S-box evaluations above; downstream byte ordering can change the displayed ciphertext but cannot restore injectivity.

6. Conclusion

MERIDIAN as specified is not a blockcipher.

References

- [1] B. Palaniswamy, P. Palmieri, and A. K. Das, “MERIDIAN: A Toroid-Inspired Permutation Block Cipher for Constrained Environments,” Cryptology ePrint Archive, Report 2026/856, 2026. <https://eprint.iacr.org/2026/856>

¹<https://www.bfswa.blog/p/llms-wont-break-symmetric-crypto>